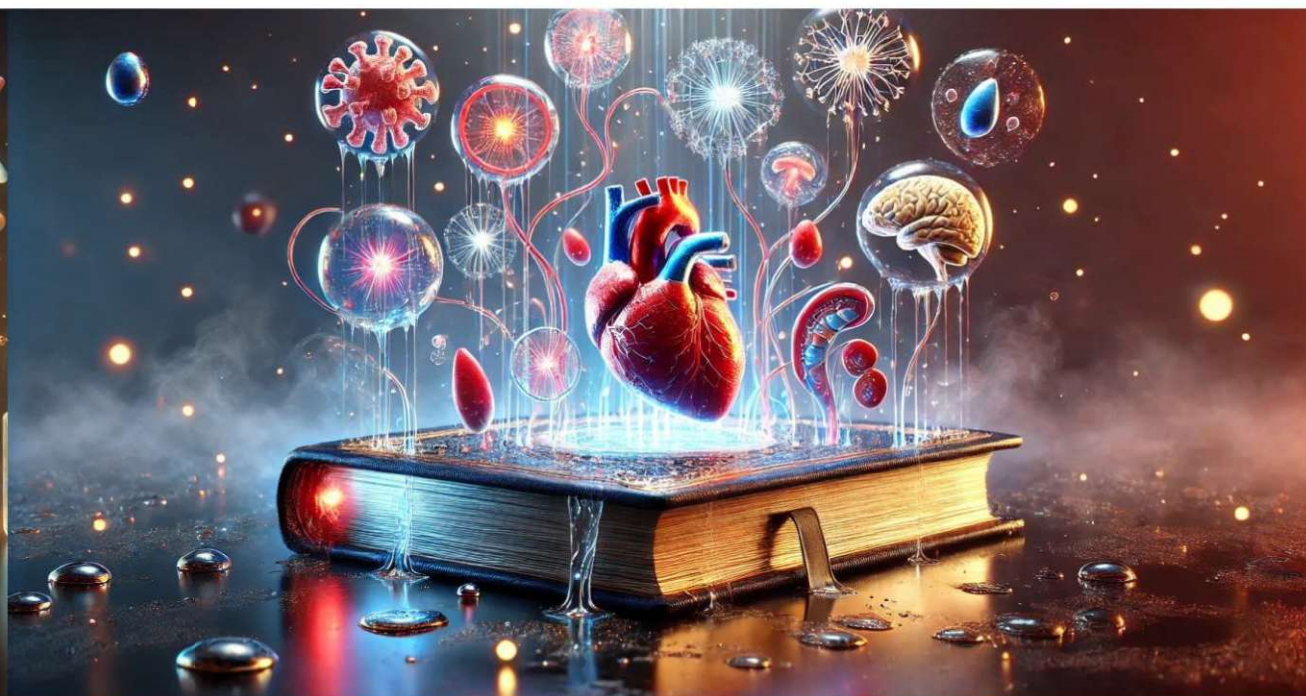
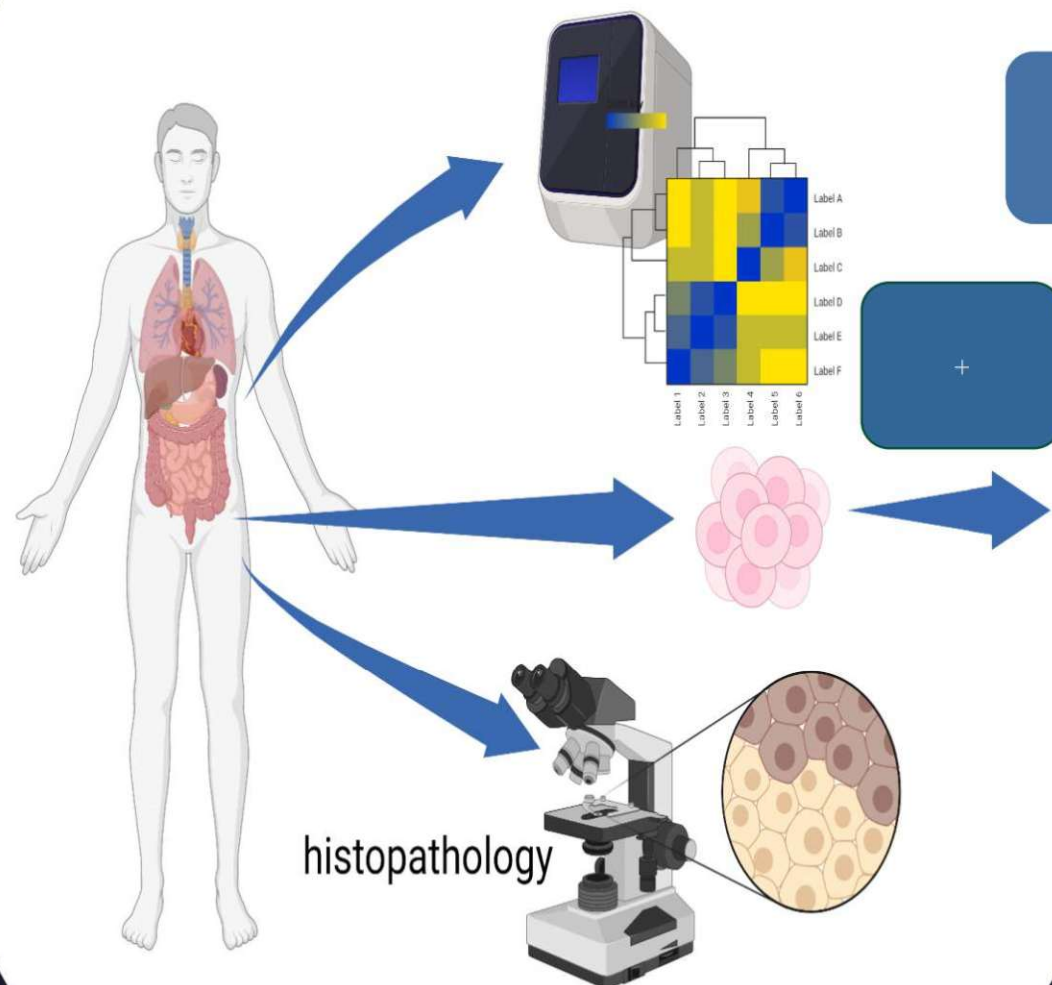




Integrated (multimodal) Omics

- Data – modeling – classification of health events
 - Health data – models – decisions or/and predictions
- Multidata type – identification of biomarkers – clinical outcomes
- Cancer cohort –intra-tumor heterogeneity – classification of data and/or events - survival estimation – etc
- Many other frames of reference

NGS analysis



National cancer registries



Central population registries

Matched by sex, age and country

> 33,160
one-year
survivors

> 212,892
comparisons

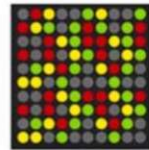
National hospital
registries
(discharge diagnoses
according to ICD to
evaluate late effects)

Multi-data type

Genomics



Transcriptomics



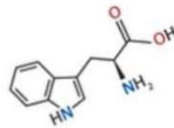
Proteomics



Epigenomics



Metabolomics



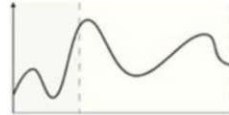
e.g., Registry data, EMR

Clinical Metadata

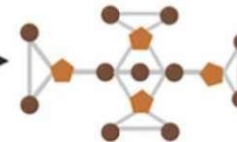


Biomarker Discovery

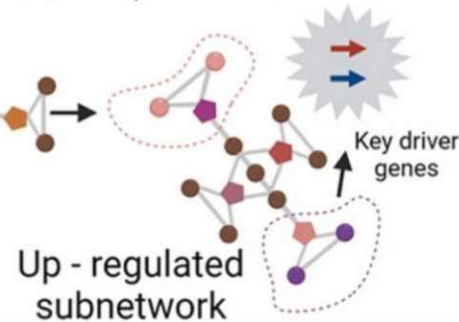
Differential Analysis



Biological Networks (GCNN, RNN, MLs)



Down-regulated subnetwork



Up-regulated subnetwork

Outcome



Deep Learning



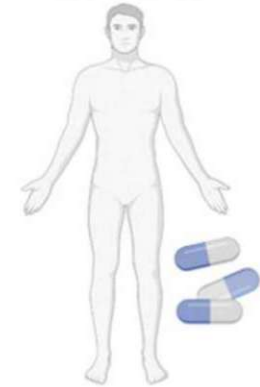
Drug A



Drug X

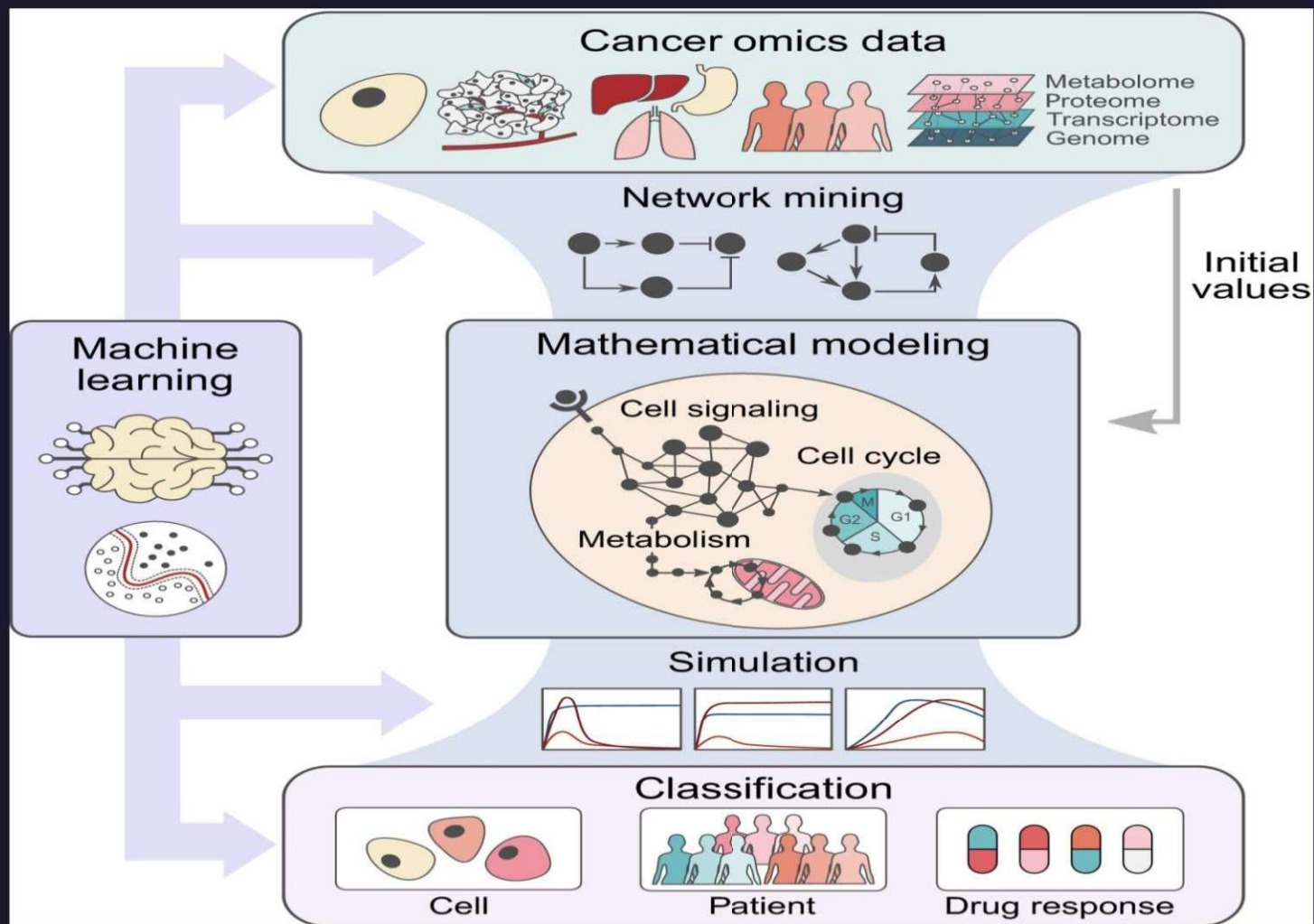
Clinical Outcome

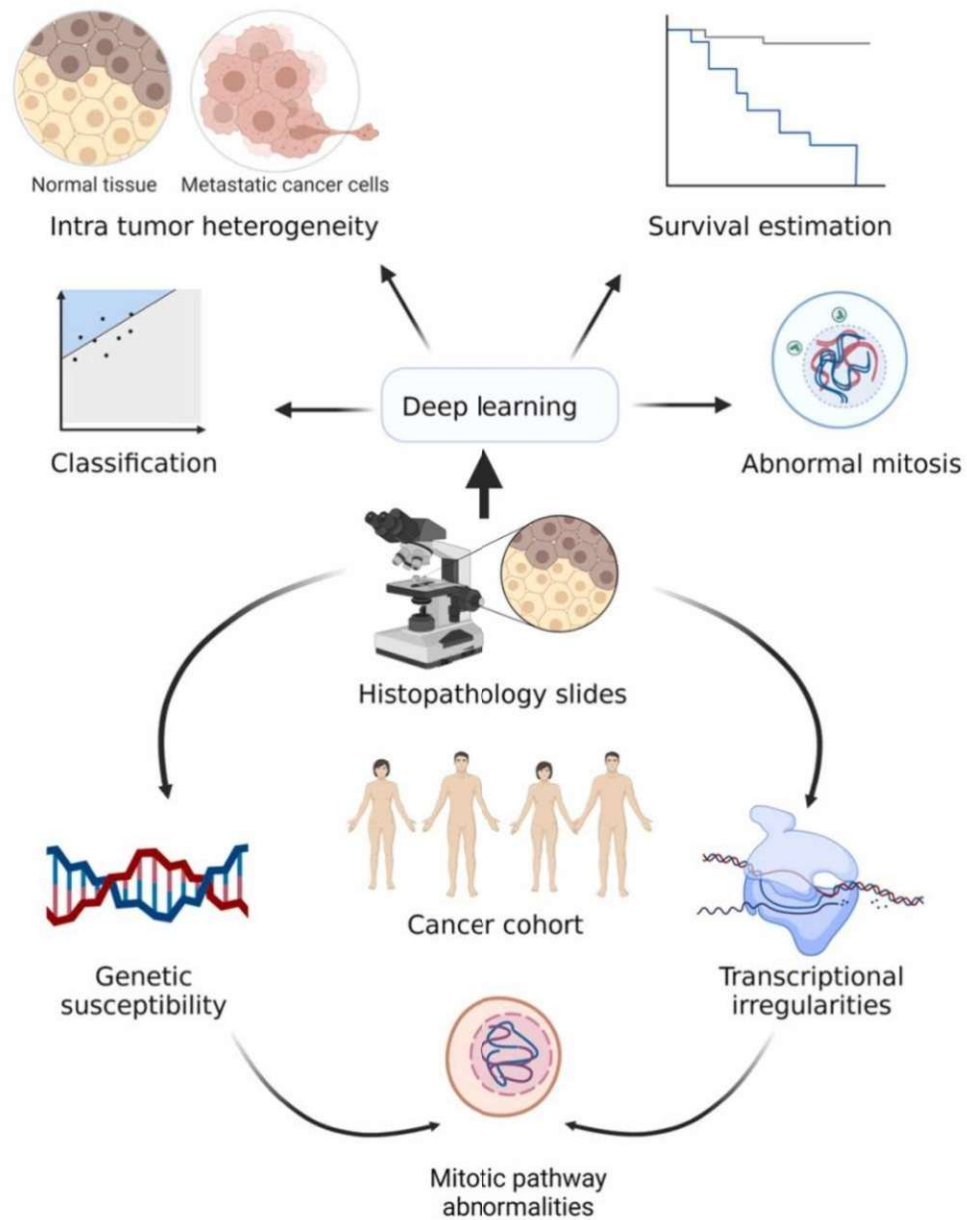
Personalized Medicine



Survival Evaluation



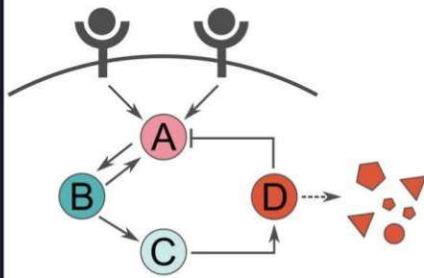




ODE MODEL

$$\frac{dx}{dt} = S \cdot v(x, \theta), x(0) = x_0$$

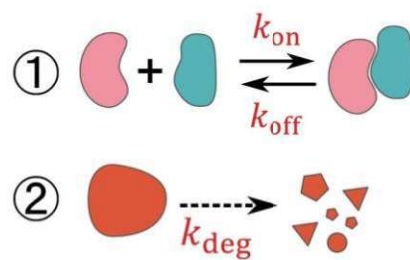
Biological network



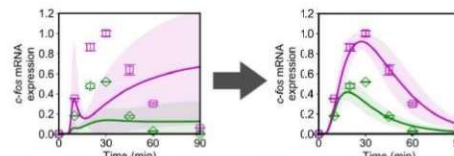
Build your own model or download from BioModels



Kinetic parameters

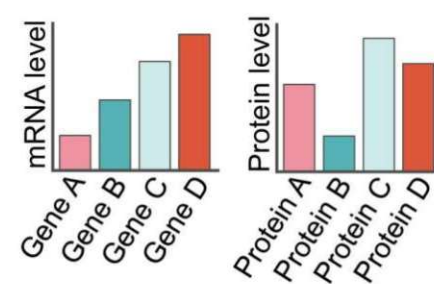


Estimate from data fitting



Fit the parameters in ODE model to data

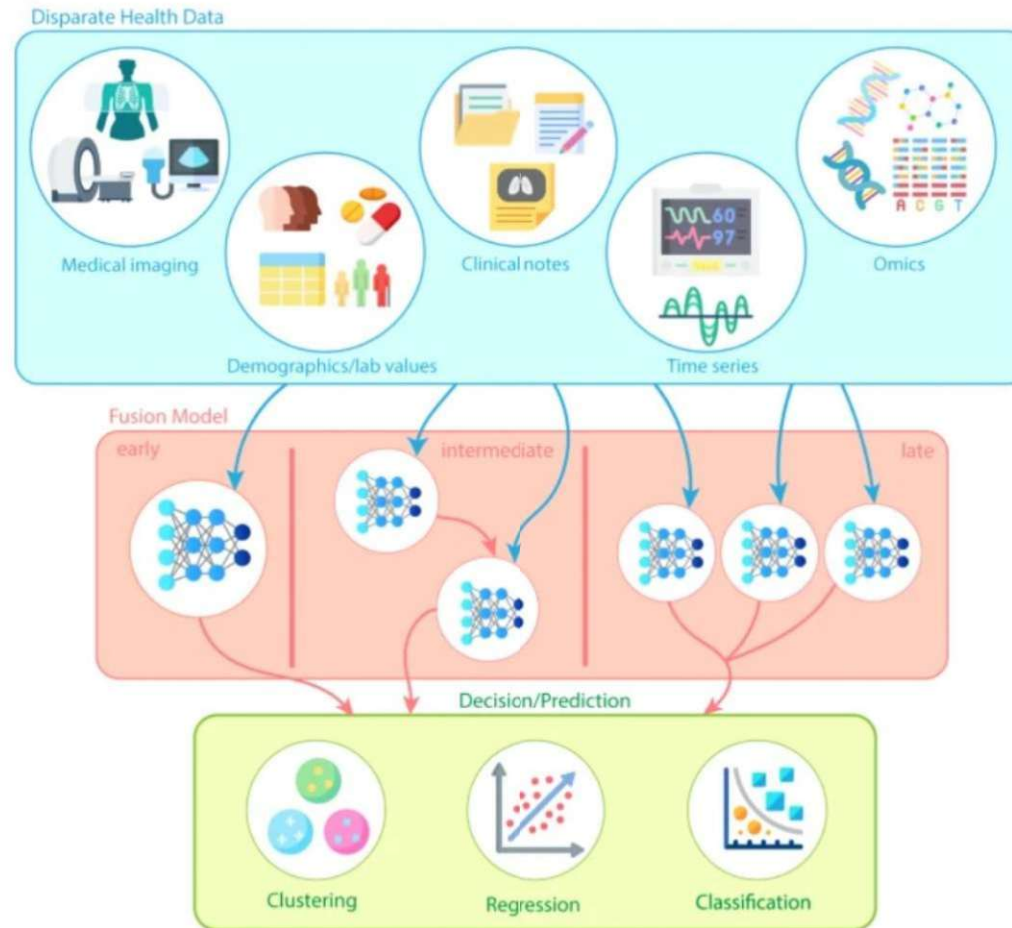
Initial conditions



Obtain from transcriptome or proteome data



Fig. 2: Early, intermediate, and late fusion; flow of information from information commons to model structure to outcomes.

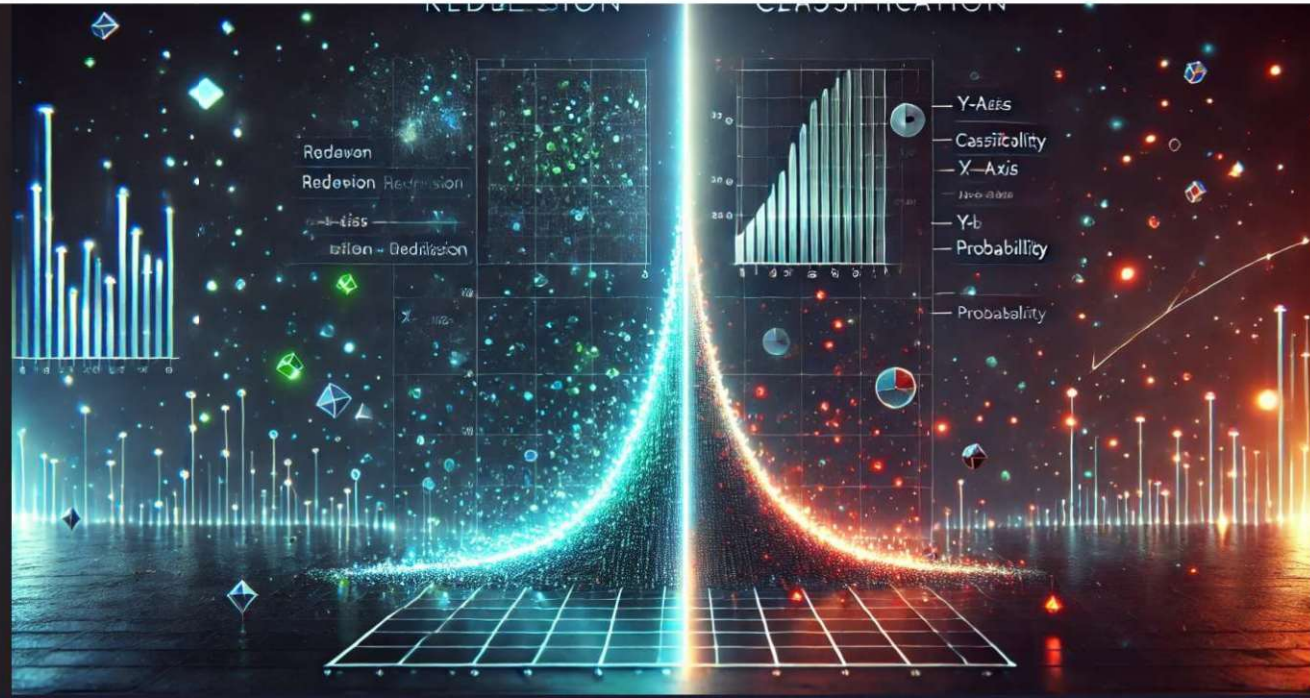


Information fusion can occur in a myriad of ways. In machine learning, early, intermediate, and late fusion is typified by if all the information flows into a single model (early), a step-wise fashion where outputs from one model become inputs for another (intermediate), and lastly, where all unique data types undergo separate modelling after which ensembling and/or voting occurs (late).



Evidence pyramid and analytics





AI is a science and a set of **computational** technologies that are inspired by — but typically operate quite differently from — the ways people use their **nervous** systems and bodies to **sense, learn, reason, and take action**



WHAT IS ARTIFICIAL INTELLIGENCE?

NEURAL NETWORKS

Computer systems designed to imitate the neurons in a brain.



NATURAL LANGUAGE PROCESSING

The ability to understand speech, as well as understand and analyze documents.

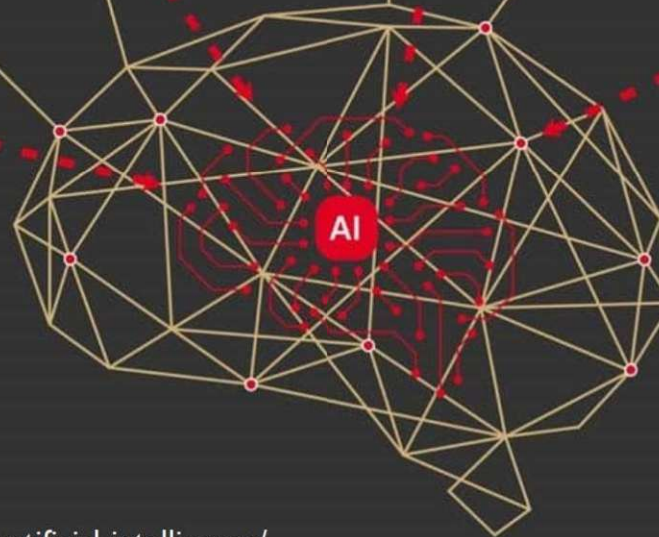


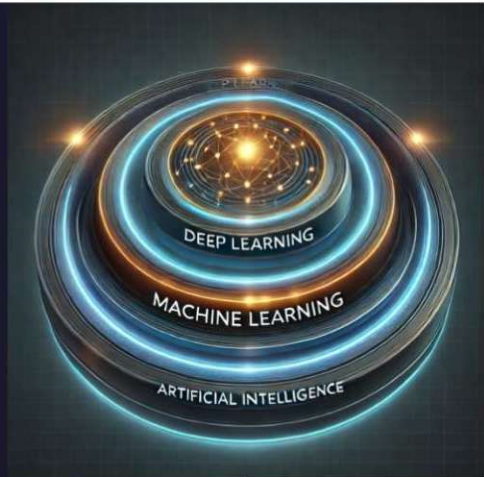
ROBOTICS

Machines that can assist people without actual human involvement.

MACHINE LEARNING

Using sample data to train computer programs to recognize patterns based on algorithms.





Artificial Intelligence (AI)

AI refers to computer systems that mimic human intelligence, enabling them to solve problems and understand language.

Machine Learning (ML)

ML is teaching systems to learn from data, enhancing performance without explicit programming.

Deep Learning

Deep Learning employs layered neural networks to find intricate patterns, excelling in tasks like image and speech recognition.



1950's



Artificial intelligence (AI)

Human intelligence exhibited by machines

1980's



Machine learning

AI systems that learn from historical data

2010's



Deep learning

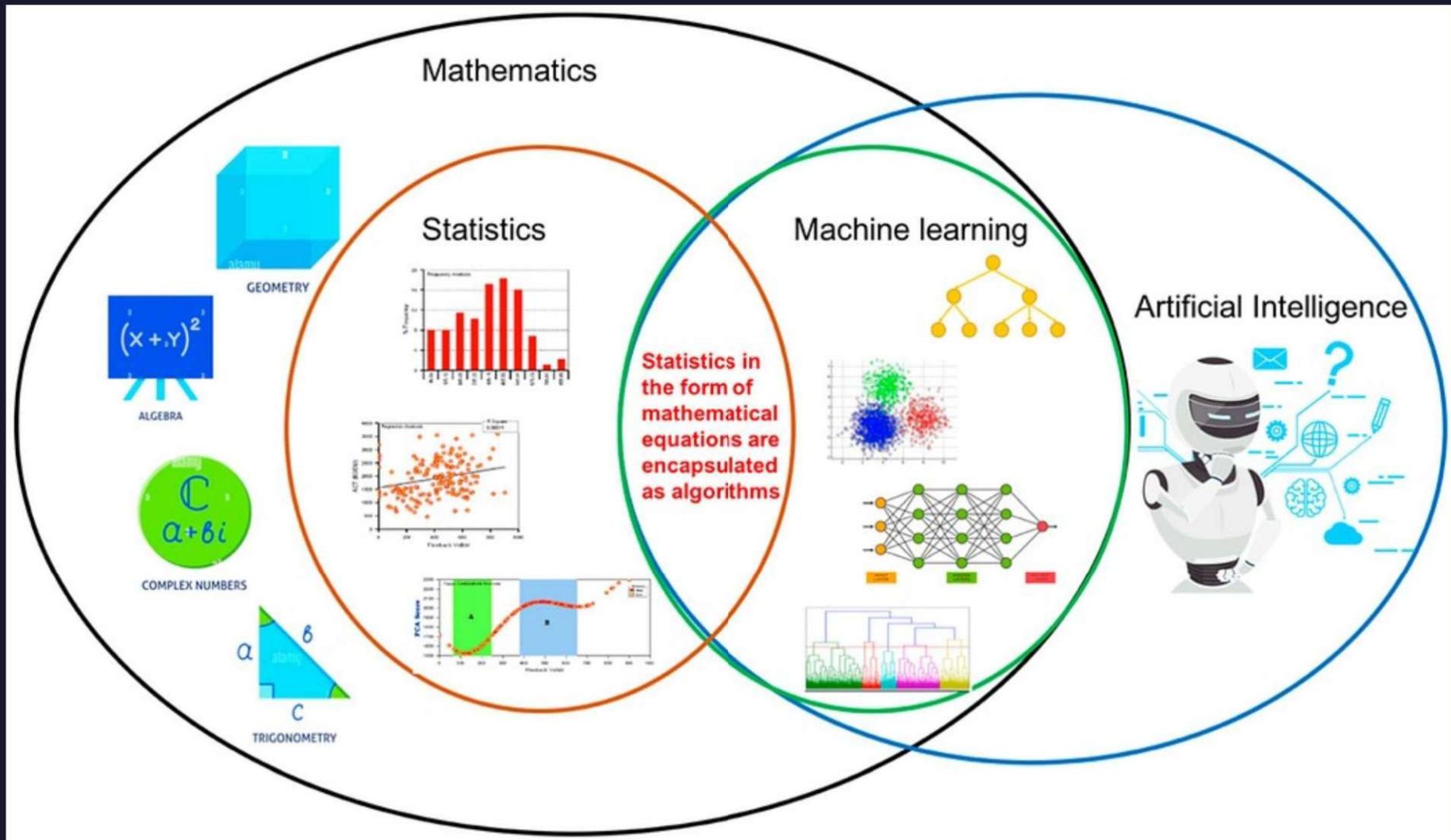
Machine learning models that mimic human brain function

2020's



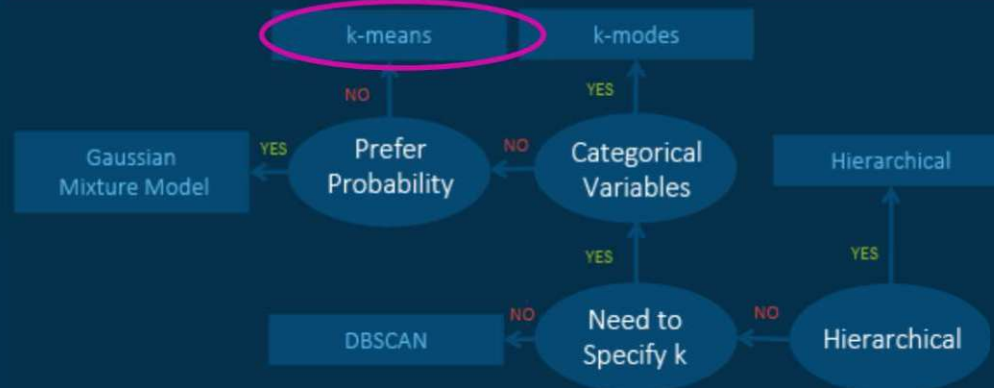
Generative AI (Gen AI)

Deep learning models (foundation models) that create original content

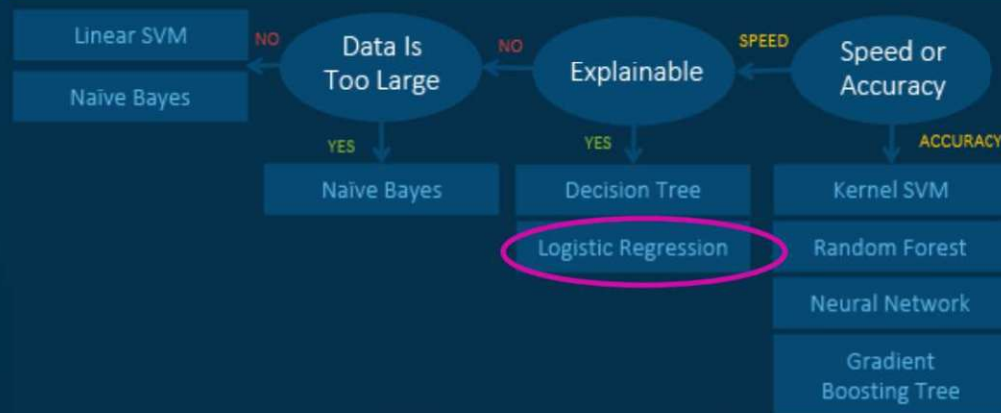


Machine Learning Algorithms Cheat Sheet

Unsupervised Learning: Clustering

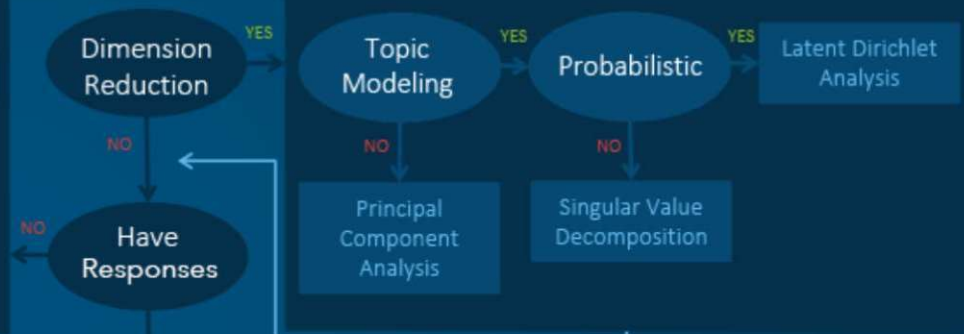


Supervised Learning: Classification

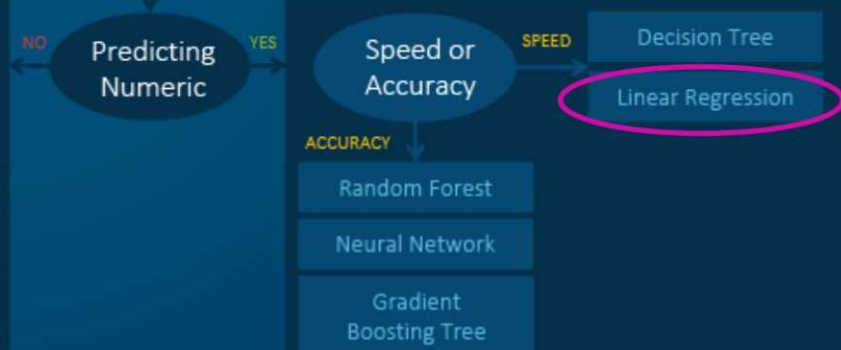


START

Unsupervised Learning: Dimension Reduction



Supervised Learning: Regression



2018-2019: Researcher, Research Unit of Chronic Conditions, CKFF

- Identify clusters of multimorbidity in the general population: a systematic **review** (PI-ongoing project).
 - Clusters of multimorbidity in the **Danish** population.
- Identify utilization of national **healthcare services** by individuals with multimorbidity in Denmark.

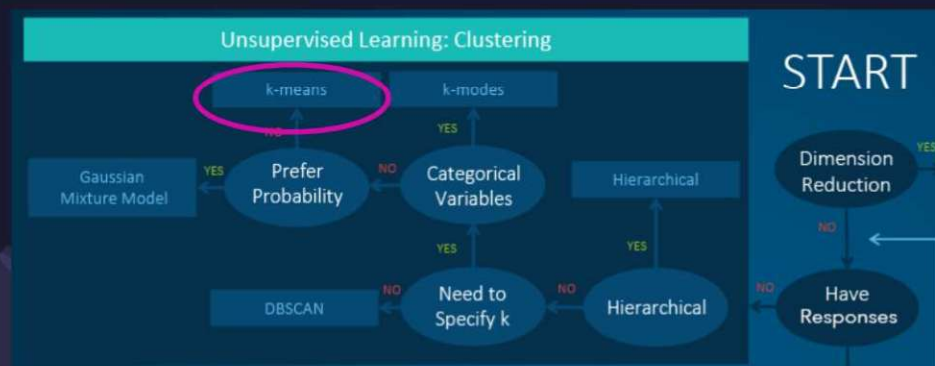
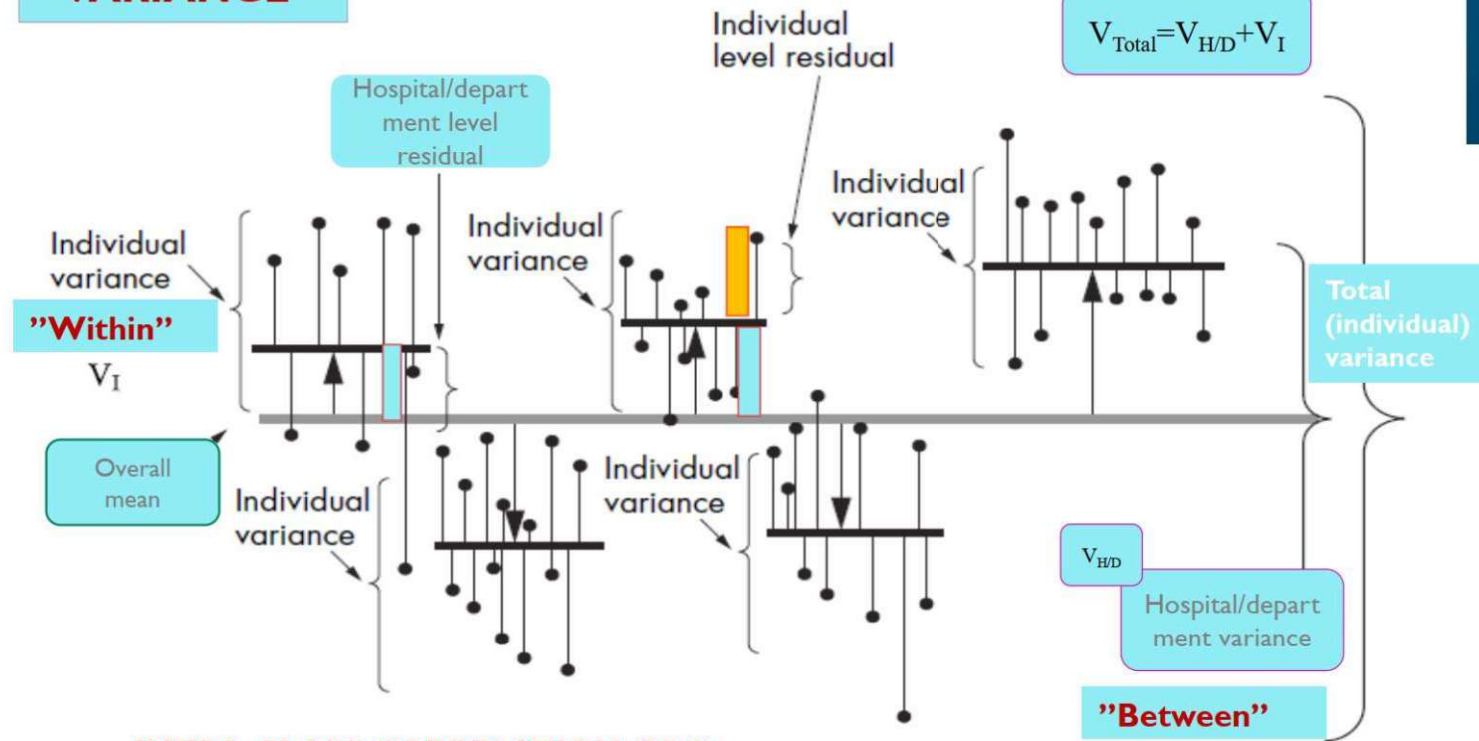


Table 1. Chronic conditions included in definition of multimorbidity (N = 16).

Allergies
Hypertension
High cholesterol
Diabetes (type 1 and type 2)
Heart disease
Stroke
Back pain
Joint disease
Osteoarthritis
Osteoporosis
Chronic obstructive pulmonary disease
Cancer
Dementia
Anxiety
Long-term use of antidepressants
Schizophrenia

<https://doi.org/10.1371/journal.pone.0214183.t001>

“VARIANCE”



INTRA-CLASS CORRELATION (ICC)

$$VPC/ICC = \frac{\text{Between } V_{\text{Hospital/department}}}{\text{between } V_{\text{Hospital/department}} + \text{within } V_I}$$

Total individual variance= within + between



Ghiith and Merlo et al (2016): <https://doi.org/10.1371/journal.pone.0148187>



Machine learning is like
statistics on steroids.



Feature	Statistics	Machine Learning
Goal	Explain relationships, hypothesis testing, and inference	Predict outcomes and optimize models for accuracy
Focus	Understanding causality , estimating parameters, deriving meaning from data	Pattern recognition, generalization , and automation
Causality vs. Correlation	causal inference (e.g., experimental design)	focuses on correlations – difficult to identify ‘why’
Data Needs	Works well with small , structured datasets	Handles massive, unstructured datasets (images, text, etc.)
Data Abundance	Structured manageable size data	Big data : video, audio, text, images,
Methods	Regression models, hypothesis testing, confidence intervals	Neural networks, decision trees, support vector machines
Complexity	Relatively simple models, interpretable results	Deep learning models with millions of parameters
Algorithmic Advances and complex data structures	struggle with high-dimensional data	ML techniques (e.g., random forests, deep learning)
Feature Engineering	Manual	Automated with e.g., Deep learning methods, convolutional networks in image recognition
Black Box Models vs. Interpretability	Interpretable	Sometimes like " black boxes "
Non-parametric vs. Parametric Models	Usually assume a distribution	often uses non-parametric
Computational power	Classical computation	GPUs , TPUs, and distributed computing

From algorithm to bedside

Evidence pyramid



Cochrane SRs Godfathers

[Archibald Leman Cochrane](#) was a Scottish physician (12 January 1909 – 18 June 1988)



- Effectiveness and Efficiency: Random Reflections on Health Services
- Adopt [randomized](#) controlled trials to support the development of less biased [clinical trials](#) and medical interventions.

[Sir Iain Geoffrey Chalmers](#) – [Cochrane Foundation](#)

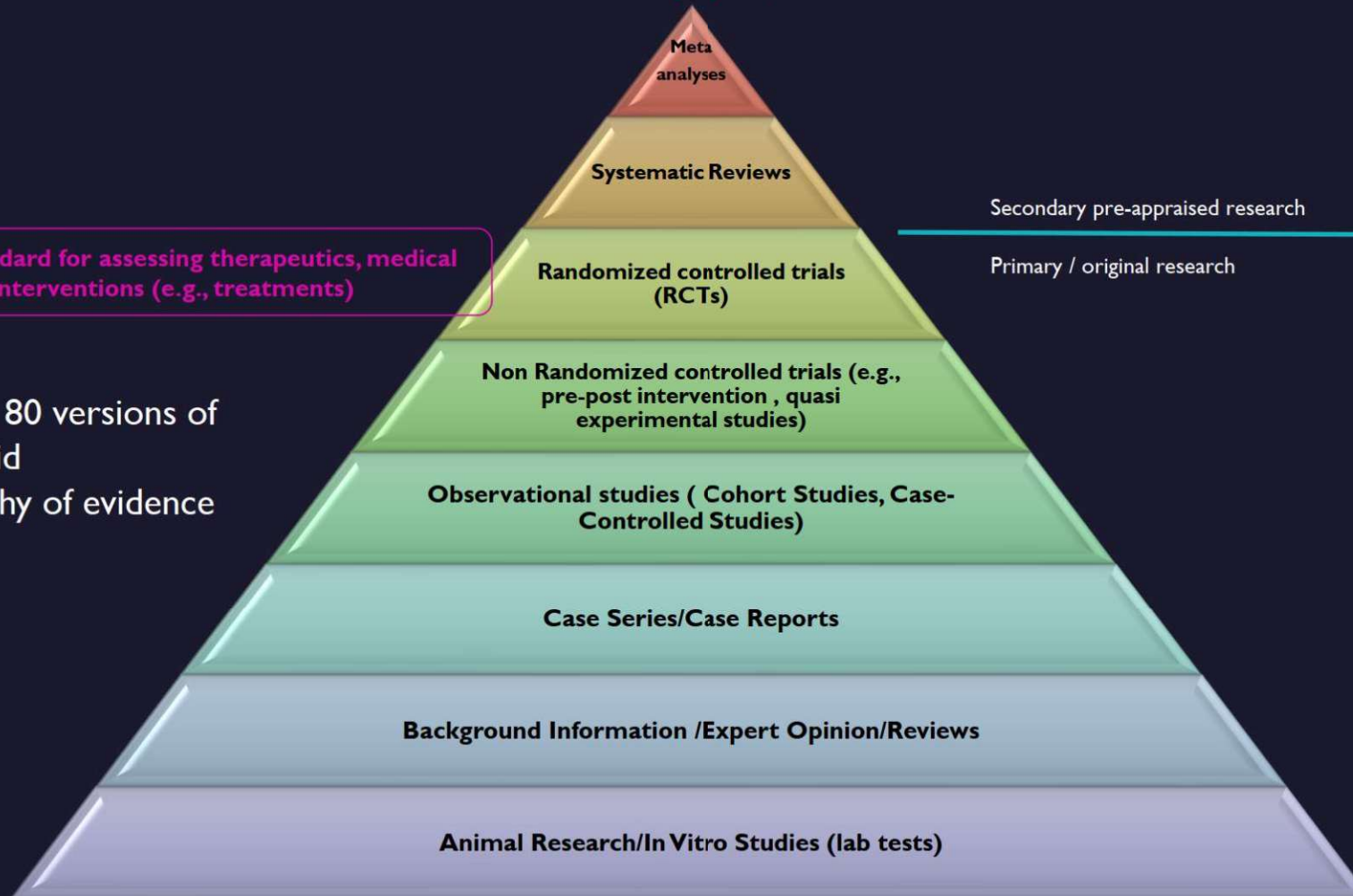


28-01-2025

@NerminGhith

40

Evidence pyramid



Gold standard for assessing therapeutics, medical interventions (e.g., treatments)

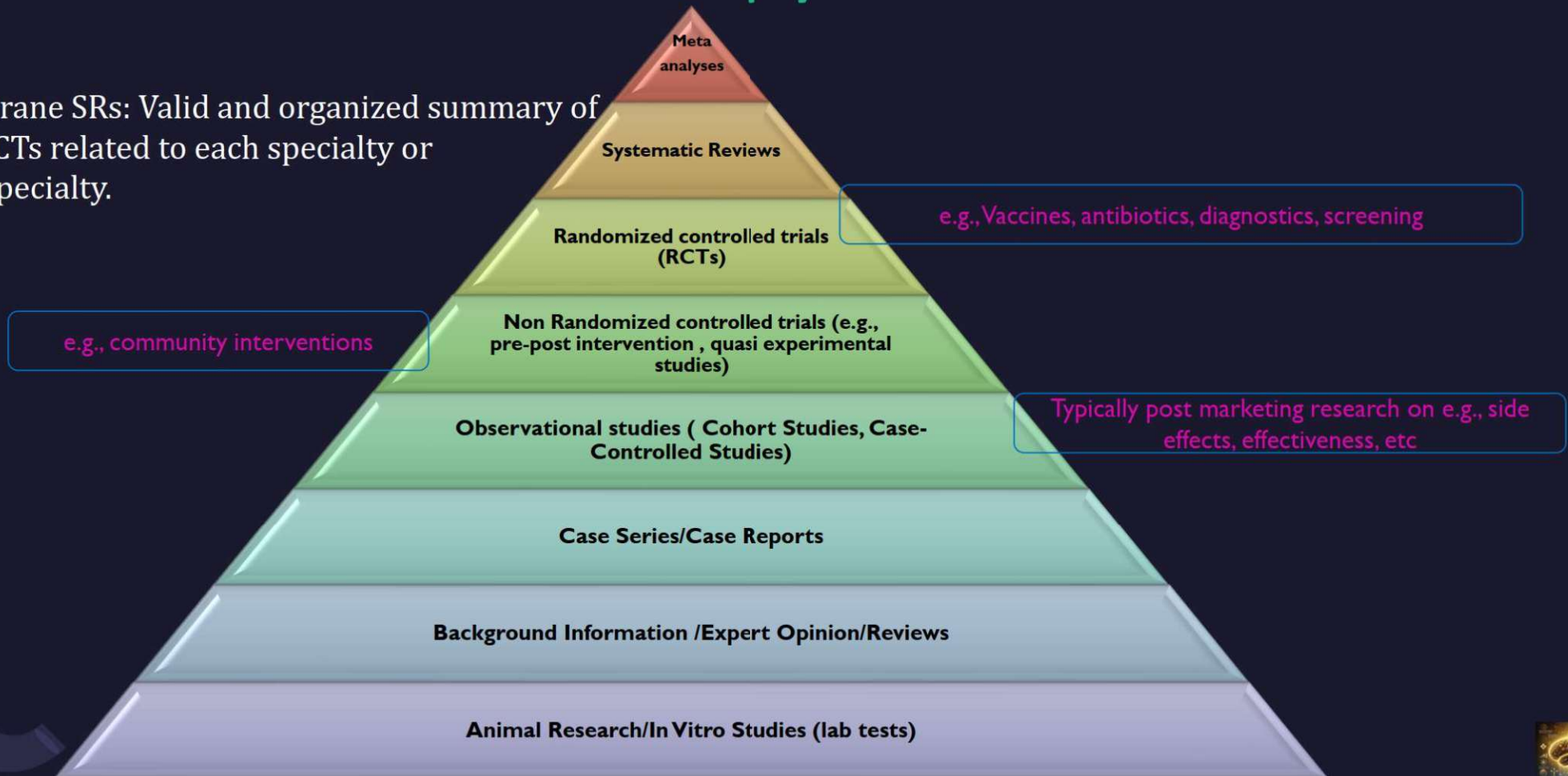
More than 80 versions of
this pyramid
Or hierarchy of evidence

Good reference on critical appraisal tools: <https://www.cebm.ox.ac.uk/resources/ebm-tools/critical-appraisal-tools>



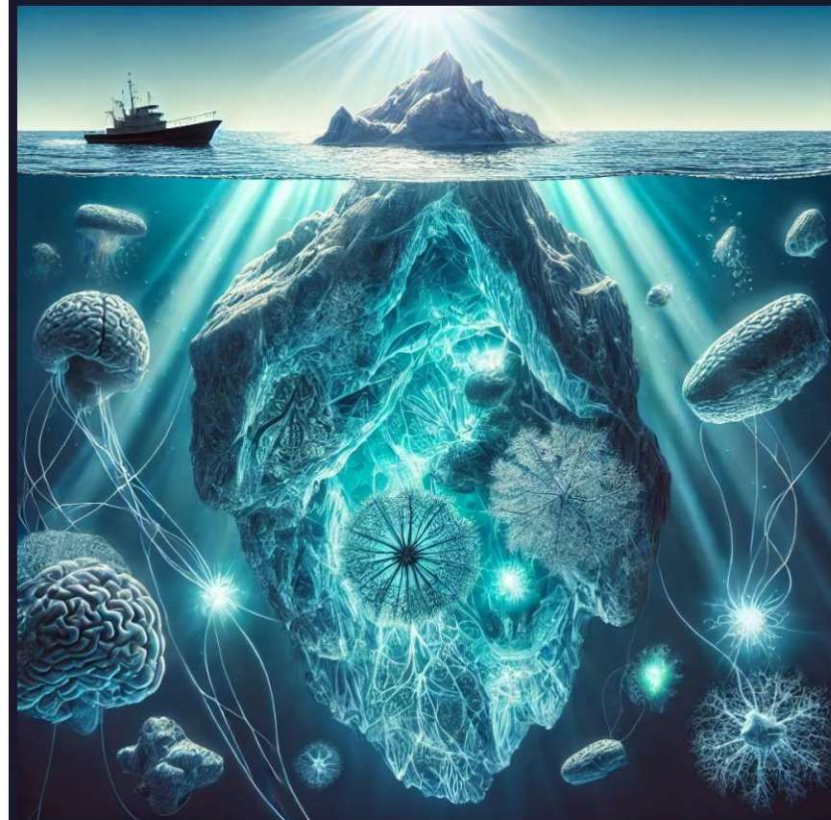
Evidence pyramid

Cochrane SRs: Valid and organized summary of all RCTs related to each specialty or subspecialty.



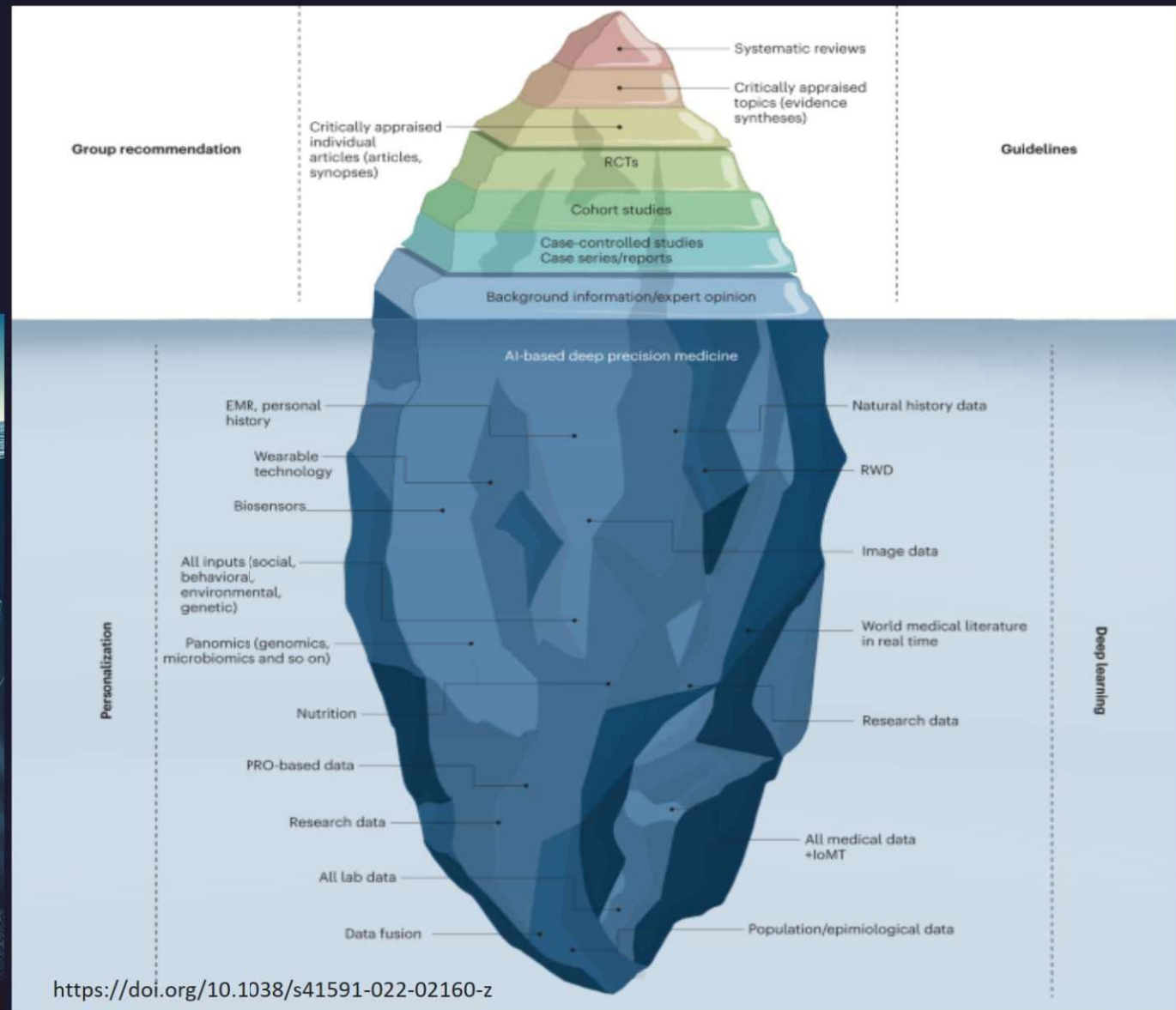


Evidence iceberg



28-01-2025

@NerminGhith





Thank You

Linkedin connect!

